A QUASI-COMPACT LOCALLY CLOSED SUBSET WITH RETROCOMPACT COMPLEMENT IN A NICE BASIS IS CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The notions of a retrocompact subset and of a constructible subset, and the hypothesis that the space has a basis of quasi-compact opens closed under finite intersections. Set up in the author's section on constructible sets, immediately before the counted statement. Fixes the hypotheses on the space and on the subset, and the conclusion.

Lemma 0.1. *Let X be a quasi-compact topological space having a basis consisting of quasi-compact opens such that the intersection of any two quasi-compact opens is quasi-compact. Let $T \subset X$ be a locally closed subset such that T is quasi-compact and T^c is retrocompact in X. Then T is constructible in X.*

Proof. Note that T is quasi-compact and open in $\overline{T}$. Using our basis of quasi-compact opens we can write $T = U \cap \overline{T}$ where U is quasi-compact open in X. Then $V = U \setminus T = U \cap T^c$ is retrocompact in U as T^c is retrocompact in X. Hence V is quasi-compact. Since the intersection of any two quasi-compact opens is quasi-compact any quasi-compact open of X is retrocompact. Thus $T = U \cap V^c$ with U and $V = U \setminus T$ retrocompact opens of X. A fortiori, T is constructible in X. □

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